

Customer Behavioral Analysis

Retail E-commerce

Gift E-commerce · Churn Prediction Pipeline

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Presentation Outline

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Business Context & Objective

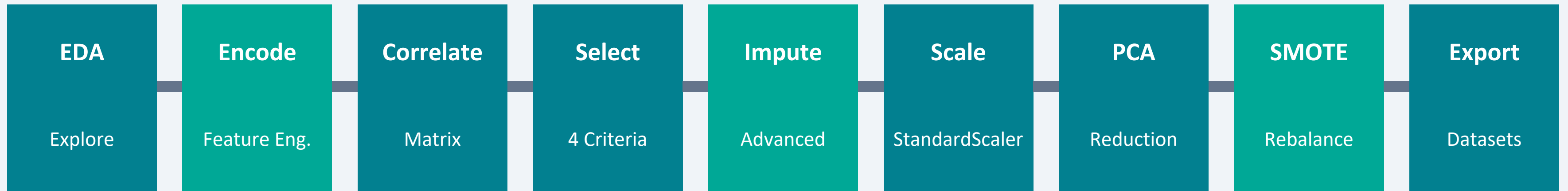
The Challenge

- A gift e-commerce company faces rising customer attrition
- 33.3% of customers have churned — 1 in 3 buyers
- Marketing budgets are wasted on lost customers
- No data-driven early warning system exists

Our Solution

- ✓ Binary churn prediction model (0 = Loyal, 1 = Churned)
- ✓ Full ML pipeline from raw data to model-ready sets
- ✓ 4,372 customers · 52 initial features
- ✓ Reproducible, leak-free preprocessing workflow

ML Pipeline Overview



Feature Reduction Journey:



Step 1 — Exploratory Data Analysis

4,372

Customers

52

Features

33.3%

Churn Rate

5

Cols w/ NaN

Missing Values by Feature

Feature	Missing %	Strategy
Age	30.34%	KNN (k=5)
SatisfactionScore	8.18%	Iterative (MICE)
SupportTicketsCount	3.00%	Median
SupportBurden	3.00%	Median
AvgDaysBetweenPurchases	1.89%	Median

Key EDA Insights

- Churn imbalance: 2:1 ratio → SMOTE needed
- NewsletterSubscribed: 100% 'Yes' → zero variance, dropped
- Sentinel values (-1, 999) found in support/satisfaction cols
- 3 date formats in RegistrationDate → parsing required
- Monetary features highly intercorrelated → redundancy

Step 2 — Encoding & Feature Engineering



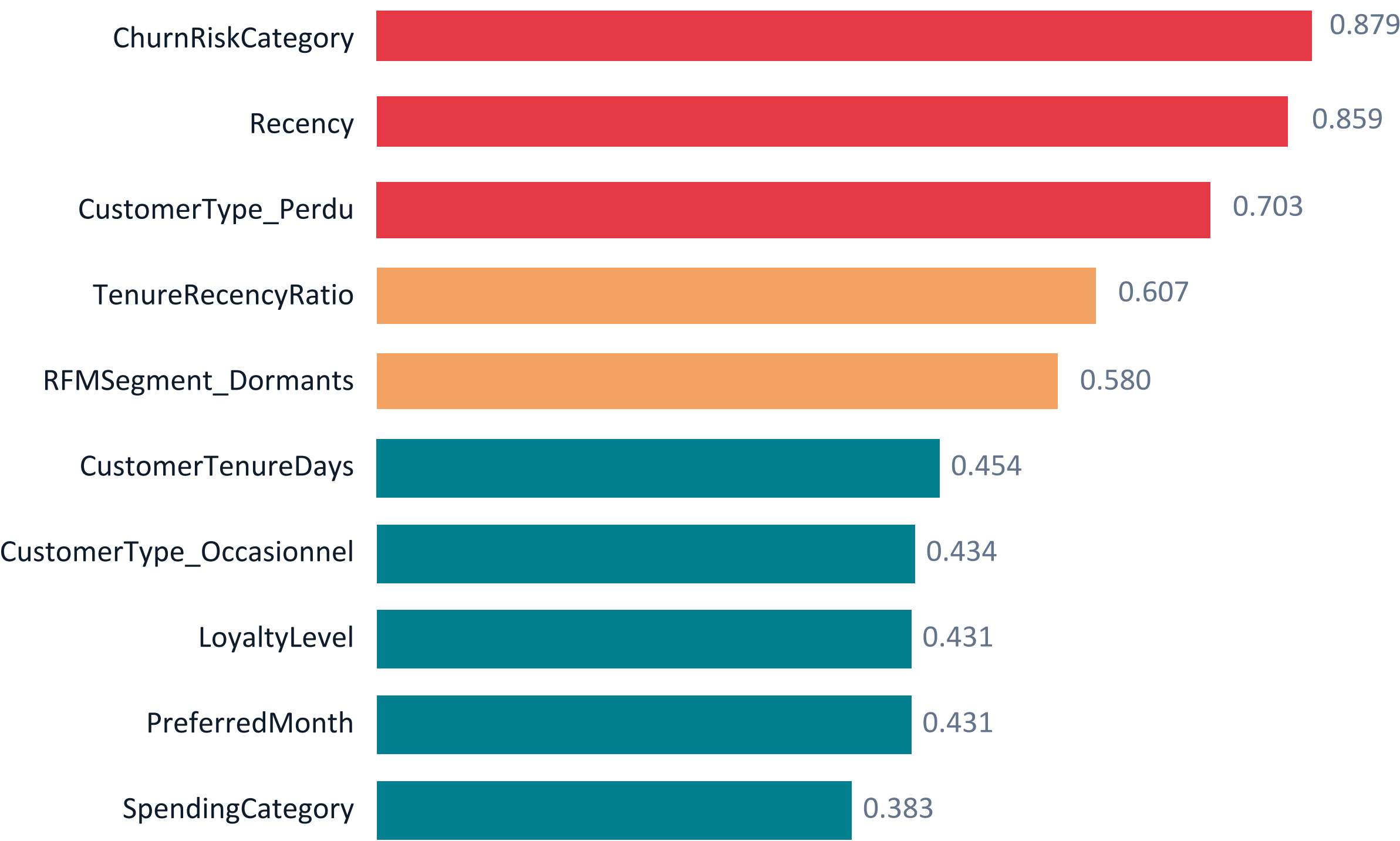
10 New Engineered Features

MonetaryPerDay	MonetaryTotal / (Recency+1) — Spending intensity
TenureRecencyRatio	Recency / (Tenure+1) — Relative inactivity
CancelPerFreq	CancelledTrans / (Freq+1) — Cancellation rate
IsNewCustomer	1 if Tenure < 90 days
HighCancelFlag	1 if Cancellation > 10%

Result: 52 raw → 87 numerical features

Step 3 — Correlation Analysis & Top Predictors

Top 10 Features Correlated with Churn



Key Insight

- ▶ ChurnRiskCategory (r=0.879) and Recency (r=0.859) are the dominant churn signals
- ▶ Recent inactivity is the strongest predictor
- ▶ 30 pairs of features exceed $|r| > 0.85 \rightarrow$ significant redundancy to remove
- ▶ Both predictors kept — they are the target-correlated, not inter-correlated

Step 4 — Feature Selection: 4 Criteria

C1	Zero Variance NewsletterSubscribed already dropped in Step 2	0 removed
C2	Missing > 50% Max NaN observed: Age at 29.99% — below threshold	0 removed
C3	High Correlation $r > 0.85$ e.g. NegativeQuantityCount $\equiv$ CancelledTrans (r=1.0)	15 removed
C4	Zero RF Importance (Gini) Regional OHE dummies with near-uniform distribution	13 removed

86 features → 71 → 58 final features selected

Step 5 — Advanced Imputation Strategy

⚠️ GOLDEN RULE:

Fit imputers ONLY on `X_train`, then apply `transform()` to `X_test`. Fitting on the full dataset would constitute data leakage.

Median Imputation

AvgDaysBetweenPurchases (1.89%)
SupportTicketsCount (3.0%)
SupportBurden (3.0%)

Few missing values, roughly symmetric distributions — median is robust to outliers

Iterative Imputer (MICE)

SatisfactionScore (8.18%)

Multivariate regression using Recency, Frequency, Tenure, SpendingCategory as context. Values clipped to [1, 5]

KNN Imputer (k=5)

Age (30.34%)

High missingness — exploits client similarity. Context: Frequency, Monetary, Tenure, Recency, AgeCategory. Values clipped to [18, 81]

Result: 0 NaN remaining in both train and test sets

Step 6 — Normalization (StandardScaler)

Why Scale?

- SVM, KNN, Logistic Regression are scale-sensitive
- PCA requires features on the same scale
- Neural networks converge faster with normalized data
- Prevents high-range features from dominating distance metrics

StandardScaler Formula

$$z = (x - \mu) / \sigma$$

Result: mean = 0, std = 1 per feature

Selective Scaling Strategy

37 continuous features

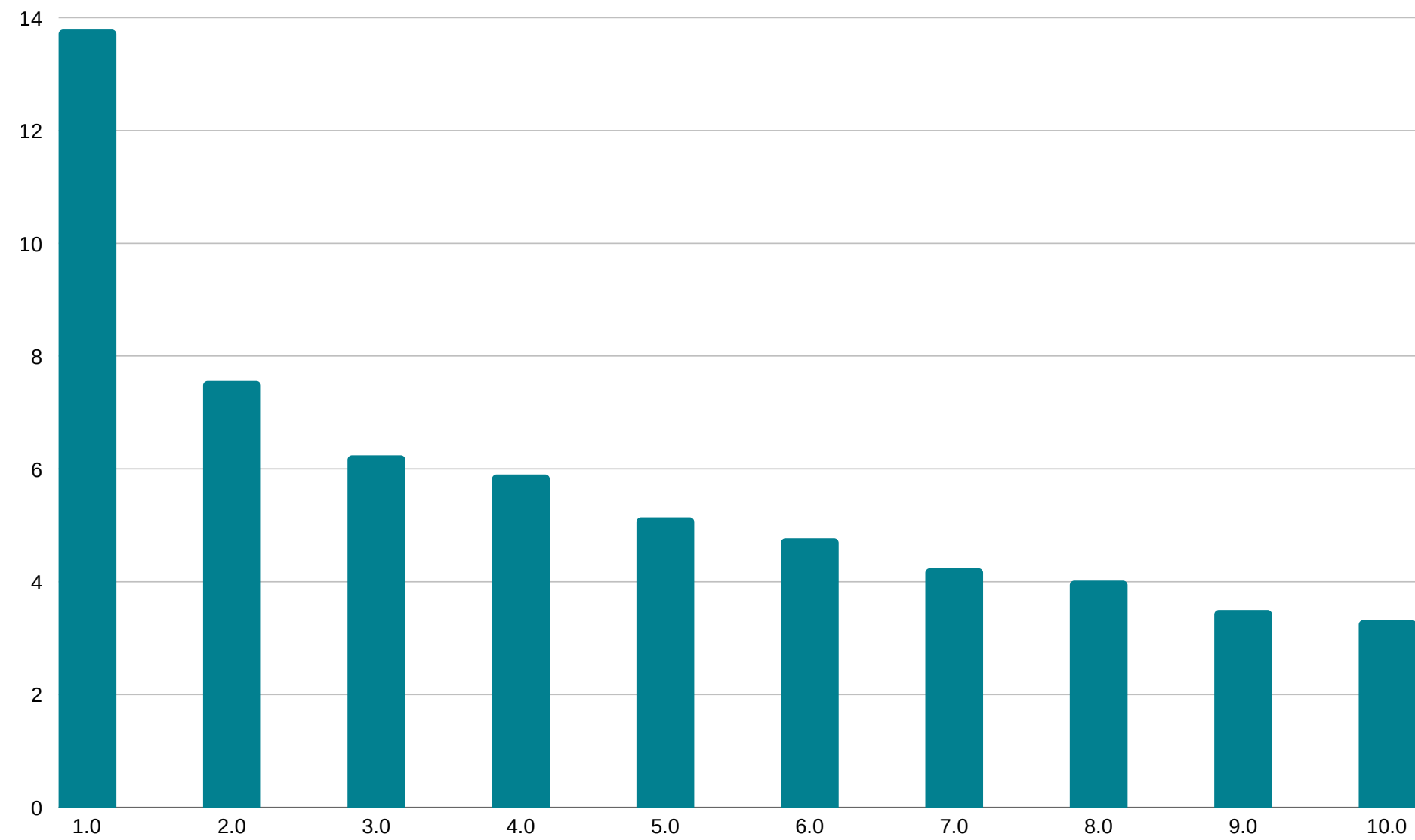
SCALED

21 binary columns {0,1}

NOT scaled

⚠ Scaler fit on X_train only — never on X_test

Step 7 — PCA: Dimensionality Reduction



3 PCA Configurations

PCA-2D: 2 comp. · 21.35% · Visualization

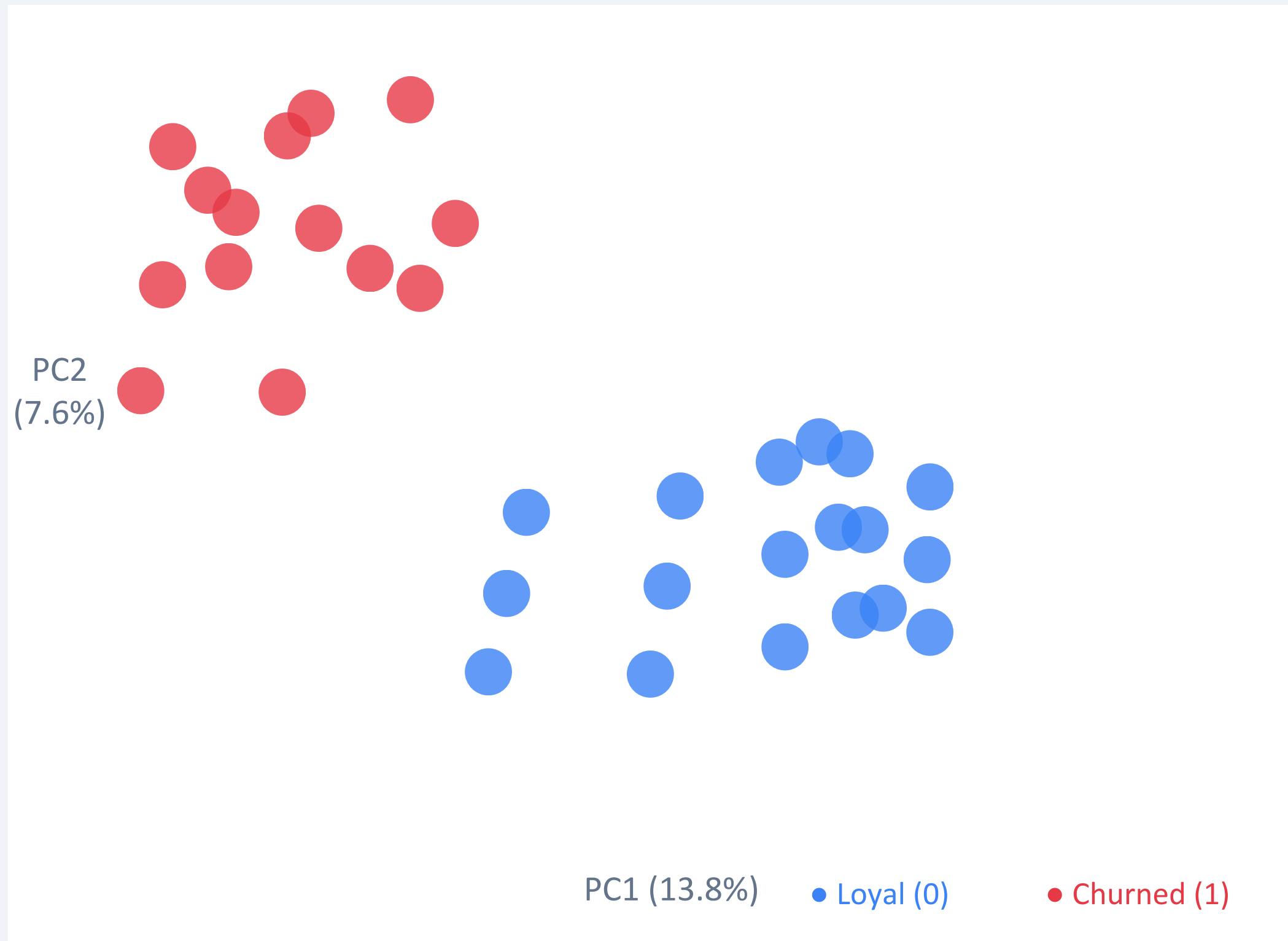
PCA-10: 10 comp. · 58.48% · Modelling

PCA-90%: 25 comp. · 90.14% · Full retention

Principal Component Interpretation

Component	Variance	Interpretation
PC1 (13.8%)	13.8%	Global engagement profile — Tenure, Spending, Frequency
PC2 (7.6%)	7.6%	Disengagement signal — Recency, ChurnRiskCategory
PC3 (6.2%)	6.2%	Spending intensity — AvgBasketValue, MonetaryAvg

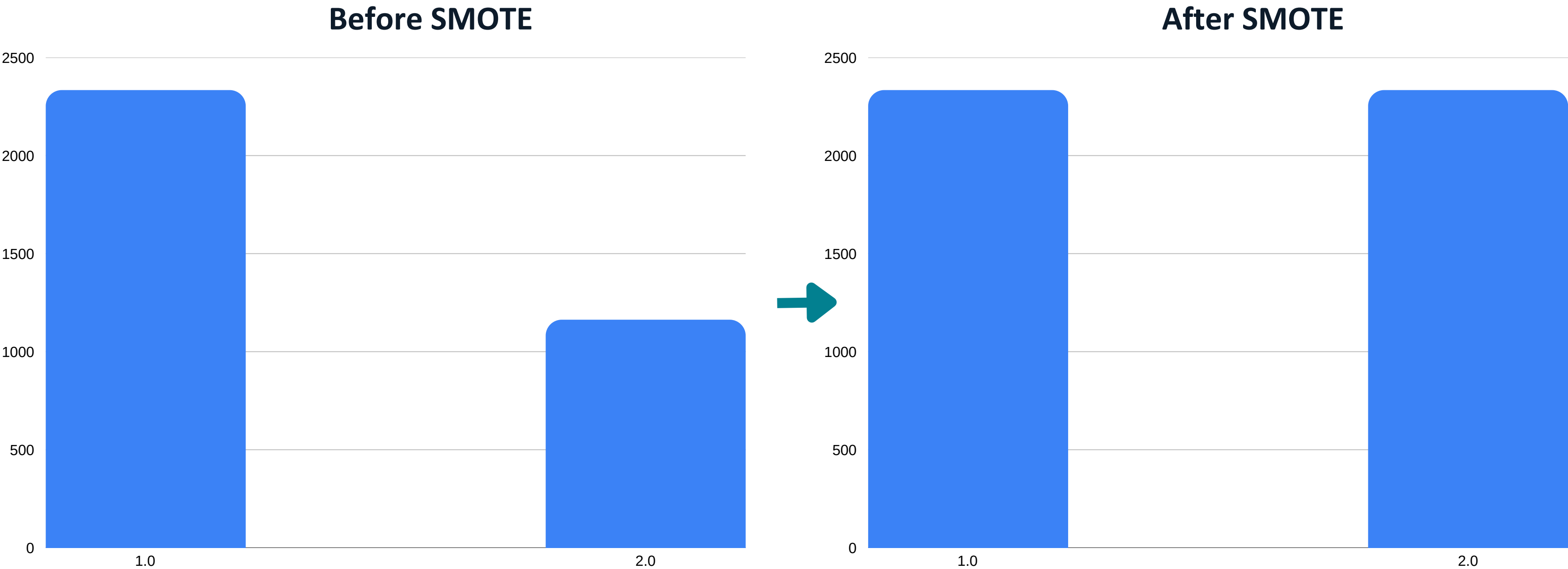
PCA — 2D Projection & Class Separability



Separability Insights

- ▶ Churned clients cluster in high-PC2 values
- ▶ PC2 is dominated by Recency + ChurnRiskCategory
- ▶ Partial separation visible even with just 2 components
- ▶ Full 58-feature space will have much cleaner separation
- ▶ Validates feature engineering quality

Step 8 — SMOTE: Class Balancing



What is SMOTE?:	Synthetic Minority Over-sampling Technique — creates new synthetic minority samples by interpolation between k nearest neighbors (k=5)
Why only on train?:	Applying SMOTE to test data would bias the evaluation — test set must reflect real-world class distribution (33.3% churn)
Result:	Train: 4,668 balanced samples (2,334 each class). Test: 875 real samples with original distribution

Step 9 — Export & Model-Ready Datasets

Dataset	Train Shape	Test Shape	Recommended For
X_scaled (58 features)	3,497 × 58	875 × 58	Logistic Regression, RF, XGBoost
X_scaled_smote (58 feat.)	4,668 × 58	875 × 58	Any classifier — balanced training
X_pca10 (10 components)	3,497 × 10	875 × 10	Fast modelling, visualization
X_pca10_smote	4,668 × 10	875 × 10	PCA + balanced
X_pca90% (25 components)	3,497 × 25	875 × 25	Compromise speed/accuracy

Saved Artefacts (joblib)

-  standard_scaler_v2.joblib — StandardScaler for continuous features
-  knn_imputer_age.joblib — KNN Imputer for Age (k=5)
-  iterative_imputer_satisfaction.joblib — MICE Imputer for SatisfactionScore
-  pca_2d.joblib · pca_10.joblib · pca_90.joblib — Three PCA models

All artefacts fit on train only → apply transform on any new data. Pipeline is fully reproducible.

Key Takeaways & Next Steps

What We Achieved:

Complete Pipeline:	9-step reproducible workflow — EDA → Export, fully leak-free
Smart Feature Selection:	86 → 58 features via 4 statistical criteria — zero information loss
Tailored Imputation:	3 strategies matched to data type and missingness level
3 PCA Configurations:	Visualization (2D), Modelling (10D), Max retention (90%)
Balanced Classes:	SMOTE on train only — test preserves real-world distribution

Next Steps:

1. Clustering (K-Means, DBSCAN)
2. Classification (RF, XGBoost, LR)
3. Flask Deployment API
4. Real-time churn scoring